Ch.9 Field Specifications

[Step 3: Create data structure]

- Is a very important step, saves you later 3x in problem fixing time
- You may ask users and management representatives to help with specifications
- Specifications help with
  1. Field integrity (data in each field is consistent and valid)
  2. Understanding data
  3. Implementation

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Three categories of field specifications

1. **General Elements:** Field Name, Parent Table, Specification Type, Source Specification, Shared By, Alias(es), Description
2. **Physical Elements:** Data Type, Length, Decimal Places, Character Support
3. **Logical Elements:** Key Type, Key Structure, Uniqueness, Null Support, Values Entered By, Required Value, Range of Values, Edit Rule, Operations Allowed

FIELD SPECIFICATIONS

General Elements

Field Name:	Specification Type: ☐ Unique ☐ Generic ☐ Replica
Parent Table:	Source Specification:
Alias(es):	
Shared By:	
Description:	

Physical Elements

Data Type:	Character Support:
Length:	☐ Letters (A-Z) ☐ Keyboard (. , / $ # %)
Decimal Places:	☐ Numbers (0-9) ☐ Special (© ® ™ Σ π)

Logical Elements

Key Type:	☐ Non	☐ Primary	Edit Rule:
	☐ Foreign	☐ Alternate	☐ Enter Now, Edits Allowed
Key Structure:	☐ Simple	☐ Composite	☐ Enter Now, Edits Not Allowed
Uniqueness:	☐ Non-Unique	☐ Unique	☐ Enter Later, Edits Allowed
Null Support:	☐ Nulls Allowed	☐ No Nulls	☐ Enter Later, Edits Not Allowed
Values Entered By:	☐ User	☐ System	☐ Not Determined At This Time
Required Value:	☐ No	☐ Yes	
Range of Values:			

Figure 9.1 *Field Specifications sheet*

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1) General Elements

General Elements

Field Name:	Employee ID Number	**Specification Type:**	☒ Unique ☐ Generic ☐ Replica
Parent Table:	Employees	**Source Specification:**	
Alias(es):			
Shared By:	Full-Time Employees, Part-Time Employees, Customers		
Description:	A unique number used to identify each employee within our organization. It is assigned during the first day of Employee Orientation and remains with the employee throughout the duration of his or her employment.		

Figure 9.4 *The General Elements category for an* EMPLOYEE ID NUMBER *field*

Table Structures

Employees	
Employee ID	*PK*
Social Security Number	
EmpFirst Name	*CAK*
EmpLast Name	*CAK*
EmpStreet Address	
EmpCity	
EmpState	
EmpZipcode	
EmpHome Phone	

Description: clear and succinct; no acronyms; no examples; no implementation information; no rephrasing of the field name; etc.

CustLast Name—the last name of a customer.

CustLast Name—the surname of a customer, whether original or by marriage, used in all formal communications and correspondence with that customer.

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What is an alias?

Table Structures

Subsidiaries

Subsidiary ID Number
Subsidiary Name
Employee ID Number
Employee ID Number
SubsStreet Address
SubsCity

Figure 9.2 *A table requiring two occurrences of the same field*

Table Structures

Subsidiaries

Subsidiary ID Number
Subsidiary Name
President ID Number
Vice President ID Number
SubsStreet Address
SubsCity

Figure 9.3 *Using aliases in place of the EMPLOYEE ID NUMBER fields*

Note: avoid aliases if possible!

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2) Physical Elements

Physical Elements

Data Type:	Numeric	**Character Support:**	
Length:	4	☐ Letters (A-Z)	☐ Keyboard (. , / $ # %)
Decimal Places:	0	☒ Numbers (0-9)	☐ Special (© ® ™ Σ π)

Figure 9.5 *The Physical Elements category for an* EMPLOYEE ID NUMBER *field*

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3) Logical Elements

Logical Elements

Key Type:	☐ Non	☒ Primary	**Edit Rule:**	
	☐ Foreign	☐ Alternate	☐ Enter Now, Edits Allowed	
Key Structure:	☒ Simple	☐ Composite	☒ Enter Now, Edits Not Allowed	
Uniqueness:	☐ Non-Unique	☒ Unique	☐ Enter Later, Edits Allowed	
Null Support:	☐ Nulls Allowed	☒ No Nulls	☐ Enter Later, Edits Not Allowed	
Values Entered By:	☐ User	☒ System	☐ Not Determined At This Time	
Required Value:	☐ No	☒ Yes		
Range of Values:	1000 - 9999			

Figure 9.7 *The Logical Elements category for an* EMPLOYEE ID NUMBER *field in table* EMPLOYEES

Note 1: PKs are marked as unique (logical); any other field is not unique, in general – see an exception example where an employee can lead only one department.

Department ID Number	Department Name	Employee ID Number
24	Shoes	445
12	Furniture	231
33	Dresses	~~445~~

Table Structures

Departments

Department ID Number
Department Name
Employee ID Number

Figure 9.6 *Should the values of* EMPLOYEE ID NUMBER *be unique? If Employee ID Number is allowed to work in (or manage) only one department, then it is unique.*

Note 2: NULL (==a blank in a record) means missing or unknown value, do NOT use it for 'None', 'Not Applicable', 'No Response' etc. ➔ include these as acceptable range of values
Use NULL, not a blank; use NULL rarely.

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General Elements: unique vs generic vs replica

FIELD SPECIFICATIONS

General Elements

Field Name:	Vendor ID Number	Specification Type:	☒ Unique ☐ Generic ☐ Replica
Parent Table:	Vendors	Source Specification:	
Alias(es):			
Shared By:	Products		
Description:	A unique number used to identify each Vendor that supplies our organization with goods or services. It is assigned when we place the first order for such goods or services with the Vendor.		

Physical Elements

Data Type:	Numeric	Character Support:	
Length:	6	☐ Letters (A-Z)	☐ Keyboard (. , / $ # %)
Decimal Places:	0	☒ Numbers (0-9)	☐ Special (© ® ™ Σ π)

Logical Elements

Key Type:	☐ Non	☒ Primary	Edit Rule:	
	☐ Foreign	☐ Alternate	☐ Enter Now, Edits Allowed	
Key Structure:	☒ Simple	☐ Composite	☒ Enter Now, Edits Not Allowed	
Uniqueness:	☐ Non-Unique	☒ Unique	☐ Enter Later, Edits Allowed	
Null Support:	☐ Nulls Allowed	☒ No Nulls	☐ Enter Later, Edits Not Allowed	
Values Entered By:	☐ User	☒ System	☐ Not Determined At This Time	
Required Value:	☐ No	☒ Yes		
Range of Values:	100000 - 200000			

Figure 9.8 *Unique field specification for the VENDOR ID NUMBER field*

Mark as **unique** (in General Elements) PKs and fields that appear only once in the entire database.

Mark as **generic** a field that is a template for other fields.

Mark **replica** for
(1) FK and or
(2) for field that is an instance of a generic field.

Required value "Yes": you cannot leave it blank, you must write something there, even NULL will do it.

"No Nulls": you are not allowed to write NULL there.

General Elements: generic vs replica

FIELD SPECIFICATIONS

General Elements

Field Name:	State	Specification Type:	☐ Unique ☒ Generic ☐ Replica
Parent Table:		Source Specification:	
Alias(es):			
Shared By:			
Description:	A state or territory within the United States in which a person, organization, or institution resides or conducts business.		

Physical Elements

Data Type:	Alphanumeric	Character Support:	
Length:	2	☒ Letters (A-Z)	☐ Keyboard (. , / $ # %)
Decimal Places:	None	☐ Numbers (0-9)	☐ Special (© ® ™ Σ π)

Logical Elements

Key Type:	☒ Non	☐ Primary	Edit Rule:	
	☐ Foreign	☐ Alternate	☒ Enter Now, Edits Allowed	
Key Structure:	☐ Simple	☐ Composite	☐ Enter Now, Edits Not Allowed	
Uniqueness:	☒ Non-Unique	☐ Unique	☐ Enter Later, Edits Allowed	
Null Support:	☐ Nulls Allowed	☒ No Nulls	☐ Enter Later, Edits Not Allowed	
Values Entered By:	☒ User	☐ System	☐ Not Determined At This Time	
Required Value:	☐ No	☒ Yes		
Range of Values:	All state abbreviations recognized by the United States Postal Service.			

Figure 9.9 *Generic field specification for a generic STATE field*

Broad

Figure 9.10 *Customized Replica field specifications for the VENDSTATE field*

FIELD SPECIFICATIONS

General Elements

Field Name:	VendState	Specification Type:	☐ Unique ☐ Generic ☒ Replica
Parent Table:	Vendors	Source Specification:	
Alias(es):			
Shared By:	State		
Description:	The state in which the vendor's headquarters are located. This data is a component of the vendor's overall mailing address.		

Physical Elements

Data Type:	Alphanumeric	Character Support:	
Length:	2	☒ Letters (A-Z)	☐ Keyboard (. , / $ # %)
Decimal Places:	None	☐ Numbers (0-9)	☐ Special (© ® ™ Σ π)

Logical Elements

Key Type:	☒ Non	☐ Primary	Edit Rule:	
	☐ Foreign	☐ Alternate	☒ Enter Now, Edits Allowed	
Key Structure:	☐ Simple	☐ Composite	☐ Enter Now, Edits Not Allowed	
Uniqueness:	☒ Non-Unique	☐ Unique	☐ Enter Later, Edits Allowed	
Null Support:	☐ Nulls Allowed	☒ No Nulls	☐ Enter Later, Edits Not Allowed	
Values Entered By:	☒ User	☐ System	☐ Not Determined At This Time	
Required Value:	☐ No	☒ Yes		
Range of Values:	CA, ID, MT, OR, WA			

More specific

When done with field specifications for all fields (yes, a lot of work, 36 fields in Mike's Bikes!), run them by representatives of users and management! Next, we create table relations…

MySQL to designate and generate PK

```
CREATE TABLE animals (
        id MEDIUMINT NOT NULL AUTO_INCREMENT,
        name CHAR(30) NOT NULL,
        PRIMARY KEY (id)
);

INSERT INTO animals (name) VALUES
        ('dog'),('cat'),('penguin’),('whale'),('ostrich’);

SELECT * FROM animals;
```

A Sample Field Specifications Form

FIELD SPECIFICATIONS

General Elements

Field Name:	**Specification Type:** ☐ Unique ☐ Generic ☐ Replica
Parent Table:	**Source Specification:**
Alias(es):	
Shared By:	
Description:	

Physical Elements

Data Type:	**Character Support:**
Length:	☐ Letters (A-Z) ☐ Keyboard (. , / $ # %)
Decimal Places:	☐ Numbers (0-9) ☐ Special (© ® ™ Σ π)

Logical Elements

Key Type:	☐ Non	☐ Primary	**Edit Rule:**
	☐ Foreign	☐ Alternate	☐ Enter Now, Edits Allowed
Key Structure:	☐ Simple	☐ Composite	☐ Enter Now, Edits Not Allowed
Uniqueness:	☐ Non-Unique	☐ Unique	☐ Enter Later, Edits Allowed
Null Support:	☐ Nulls Allowed	☐ No Nulls	☐ Enter Later, Edits Not Allowed
Values Entered By:	☐ User	☐ System	☐ Not Determined At This Time
Required Value:	☐ No	☐ Yes	
Range of Values:			